

[Final Exam (Sample)]

Spring 2024: CSE 309: Theory of Automata

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Sunday, May 26, 2024. Total Marks: 35 (final) + 5 (quiz) + bonus, Duration: 3 hours

Name: _____, Student ID: _____

Please read these instructions carefully:

- This exam contains 8 questions. Question 1 contributes towards the Quiz component of the course. The rest of the questions contribute towards the final exam component. Question 8 is a bonus question.
- Use of calculators, mobile, or any other communication devices is strictly prohibited for the entirety of the exam period.
- Please submit your devices in your bag at the front of the examination room.
- You may keep writing material with you on your desk.
- Read all questions carefully before you start answering and prioritize which questions to attempt first.
- You may ask for extra sheets for your rough work, scribbling, future planning etc., **anything on extra sheet(s) will not be graded.**
- Please write your name and student ID on this question paper clearly.
- Please do not use pencils or red/green/yellow pens for answers that you want to be graded.
- Instructor(s) will not entertain any queries during the exam.
- Please attach all extra sheets with your answer book.
- Please write clearly and legibly and avoid overwriting.
- Pay attention to the instructions for each question.
- Acquisition of answers through unfair means will automatically cancel your exam.
- Keep track of the time.
- **Your answer(s) will not be graded if unreadable (by the instructors).**
- **Your exam will not be graded if you don't follow these instructions.**

— Good Luck —

Question 1: Quiz 5 marks

Consider the following pair of languages:

$\text{DIAG} := \{ \langle M, k \rangle : M \text{ is a square matrix and sum of values on its diagonal is at most } k \}$

$\text{TSP} := \{ \langle M, k \rangle : M \text{ is a square matrix representing inter-city distances}$
and there is tour that visit all cities of total cost at most $k \}$

Note that TSP is a known **NP**-hard problem, whereas DIAG is in **P**. Which of the following must be true, assuming $\mathbf{P} \neq \mathbf{NP}$? [No justification is needed.]

- (a) False TSP $\in$ **P**
- (b) True DIAG is in **NP**
- (c) True There is a polynomial-time reduction from SAT to TSP.
- (d) True There is a polynomial-time reduction from DIAG to TSP.

(a):
TSP is NPH which is disjoint from set NP (P is inside NP) so an NPH problem can never be in P

(b):
DIAG is in P which is a subset of NP. Everything in P is also in NP.

(c):
SAT is an NP problem and since TSP is an NPH problem then all NP problems are reducible to that NPH problem.
hence SAT is reducible to TSP.

(d):
DIAG is a P problem, which is also an NP problem, hence all NP problems are reducible to an NPH problem.

Question 2: *6 marks*

- (a) (3 marks) Fix the alphabet $\Sigma = \{0, 1\}$. A *run* in a string $w \in \Sigma^*$ is a maximal substring of consecutive identical characters. For example, the string 001110011 has exactly four runs: $001110011 = 00 \cdot 111 \cdot 00 \cdot 11$.

Let L be the set of all strings in Σ^* that contains only odd length runs. For example, L contains 000111111110 but does not contain the string 10000111 . Describe a regular expression for L .

$$1.(11)^*.(0.(00)^* \cup 1.(11)^*)^* \quad \cup \quad 0.(00)^*.(0.(00)^* \cup 1.(11)^*)^*$$

but problem is, it accepts 0011 so its wrong, cant find a better RE.

- (b) (3 marks) Describe a CFG that generates the following language:

$$L = \{a^x b^y c^z : y = x + z \text{ and } x, y, z \geq 0\}$$

$S \rightarrow aXbbYc \mid \text{epsilon}$

$$X \xrightarrow{\epsilon} aXb$$
$$Y \rightsquigarrow bYc \mid \text{epsilon}$$

Question 3: 6 marks

Indicate whether following statements are **true** or **false** also justify your answer i.e., give a simple proof for a **true** statement and a counterexample for a **false** statement.

- (a) True Every regular language is context-free.



Proof 1:

The class of RL is a subset contained within the class of CFGs. Hence every RL is in the class of CFGs.

Proof 2:

For every RL we can create an FA (DFA/NFA). Each DFA can be made into a CFG using the format:

For each state in the DFA, introduce a non-terminal symbol in the CFG. For the start state of the DFA, make the corresponding non-terminal the start symbol of the CFG. For each transition in the DFA of the form $q_i \xrightarrow{a} q_j$ where q_i and q_j are states and a is an input symbol, add a production rule in the CFG of the form $A_i \rightarrow aA_j$, where A_i and A_j are the non-terminals corresponding to states q_i and q_j respectively. For each accepting state q_f in the DFA, add a production rule in the CFG of the form $A_f \rightarrow \epsilon$ where A_f is the non-terminal corresponding to q_f and ϵ represents the empty string. Since every regular language can be recognized by a DFA, and we can construct a CFG from any DFA, it follows that every regular language is context-free.

- (b) False Following language L is **NP**-hard.

$$L = \{ \langle M \rangle : M \text{ is a DFA that accepts all strings} \}$$

so we make a TM S which decides the language L in polynomial time. what S does is:

"on input $\langle M \rangle$,

1. construct a DFA D from $\langle M \rangle$, if not a valid DFA, reject.
2. reverse the states of D , to generate DFA B . Now B is the complement of the language(M).
3. Run Edfa on $\langle B \rangle$ if it accepts, accept. else reject."

Each stage of S runs in polynomial time. Hence S is in P , which is in NP . Anything in NP cannot be in $NPHard$. Hence False.

Now if we assume L is in NPH then that means every language in NP reduces to L . But by above proof L is in P , which then means that every language that reduced to L would also be in P (by theorem) which then proves $P=NP$ which is wrong. It is not certain hence a contradiction that L is in NPH .

Question 4: 5 marks

Show that if A is *recognizable* and $A \leq_m \bar{A}$, then A is *decidable*.

before reading this proof, remember a theorem: "A language is decidable if and only if it is Turing Recognizable and Co-Turing Recognizable."

If language A is reducible to language B then the language $A^c/\bar{A}$ (complement of A) is also reducible to language $B^c/\bar{B}$ (complement of B). this is because if A is reducible to language B then the complement (rest of A) should also be reducible to the complement (rest of B).

SO in this question, if A is reducible to $\bar{A}$ then $\bar{A}$ is also reducible to $(\bar{A})^c$ which is A . so, $A \leq_m \bar{A}$ AND $\bar{A} \leq_m A$.

So using the second reduction, and the theorem "if A reduces to B and B is in P , then A is in P . this is applicable for all B properties, A will have that property" then since A is Turing Recognizable then $\bar{A}$ is also Turing Recognizable.

Now using the above mentioned theorem in Red, now we have proved that A complement is also Turing Recognizable and it is given that A is also TR, hence A is decidable.

Question 5: *6 marks*

Show that following problems are in class **NP**.

- (a) **3SAT**: Given a CNF formula ϕ with n variables and m clauses, is there an assignment to the variables that satisfies all the clauses.

We can prove this is in NP using a solution/certificate C that is verified in polynomial time. Suppose we are given a certificate C of the solution of 3SAT. We create a DTM M which verifies the certificate C :

"on input C which has the boolean formula f_i , with its truth assignment for n variables:

1. run this loop for each of the m clauses from $i = 0, 1, \dots, m$
2. get each clause and replace each variable in the clause with their truth assignment.
3. if any of the variables in the clause is 1, increment i and repeat from step 1 again for the next clause.
4. if all of the variables in the clause is 0, reject the solution.
5. if all the clauses are checked and it has not been rejected. uptill now, that means all the clauses equal to 1 and f_i is equal to 1. the given certificate is correct. accept."

Thus, a certificate C is verified in polynomial time as m, n are finite natural numbers and $n \cdot m$ is a finite number. so it is contained in NP class.

- (b) **Dominating Set:** Given a graph $G = (V, E)$ and a number k , is there a set $V' \subseteq V$ such that $|V'| \leq k$ and for every vertex $v \in V$, either v is in V' or v is adjacent to a vertex in V' .

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Question 6: *5 marks*

If L is a language over some alphabet Σ^* then the complement of L denoted as $\overline{L}$ is defined as $\overline{L} = \Sigma^* - L$. We say a language L is in the class **coNP** if its complement $\overline{L}$ is the class **NP**. Prove/disprove that “**NP** = **coNP** if and only if $A \leq_P B$ and $B \leq_P A$ where A is an **NP**-complete languages and B is a **coNP**-complete language.”

Question 7: *7 marks*

Let $T = \{\langle M \rangle \mid M \text{ is a TM that accepts } w^{\text{rev}} \text{ whenever it accepts } w\}$. Prove by reduction that T is undecidable.

Please ensure that you give a complete argument regarding why your reduction works. You can choose any undecidable language that we have seen in class for your reduction.

Question 8: Bonus 5 marks

Show that the class **NP** is closed under the star operation. That if L is in **NP** then L^* is also in **NP**.

Proof 1:

We have a TM M which decides the language L . We create a TM N which decides the language L^* .

N does:

"on input string w :

1. run a loop from $i = 0, 1, \dots, n$ where n is the length of w
2. non deterministically, break w into i parts such as $w = w_1, w_2, \dots, w_i$
3. for each w_j broken from $j = 0, 1, \dots, i$
4. run w_j on M . if M accepts, move forward, if M rejects, stop this j loop and move to the next i .
5. if all w_j are accepted on M , accept.
6. reject if not accepted after i is wrong till n .